Estimation of Cluster-Specific Causal Effects on Spatially Associated Survival Data Using SoftBART

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The Problem: Treatment Delay in Breast Cancer

Does delaying breast cancer treatment (>90 days) cause worse survival? How does the effect vary across geographic regions?

Breast Cancer Facts

Most diagnosed cancer in US women
2nd leading cause of cancer death
Treatment Delay (TD): diagnosis to first treatment
TD > 90 days ⇒ worse 5-year survival

Florida Cancer Registry (FCR)

$N = 68{,}009$ patients in $K = 67$ counties
41.3% right-censored
Treatment: $Z = 1$ (TD $\le$ 90 d), $Z = 0$ (TD > 90 d)
Confounders: Age, HR status, Tumor Grade, Biopsy Delay

Key Challenges

Data Challenges

- **Spatial dependence** among neighboring counties
- **Right-censoring** (41.3%)
- **Confounded treatment** — TD not random
- **Clustered** patients within counties

Modeling Challenges

- **Non-linear** treatment–confounder–outcome relationships
- **Unmeasured confounding** at county level
- Need **county-** and **patient-specific** effects
- Existing methods: no spatial + nonparametric

Our Innovation

PS-LND: First framework combining **nonparametric causal inference** (SoftBART) with **spatial modeling** (DAGAR) for clustered survival data

- ✓ **Flexible:** SBART captures smooth non-linear effects without pre-specifying interactions
- ✓ **Proper spatial:** DAGAR guarantees valid distribution with interpretable $\rho \in (0, 1)$
- ✓ **No feedback:** two-stage design isolates PS from outcome model
- ✓ **Bayesian UQ:** full posterior with credible intervals
- ✓ **Multi-level:** patient, county, and state-wide causal estimates
- ✓ **Robust:** works under misspecification & distributional violations

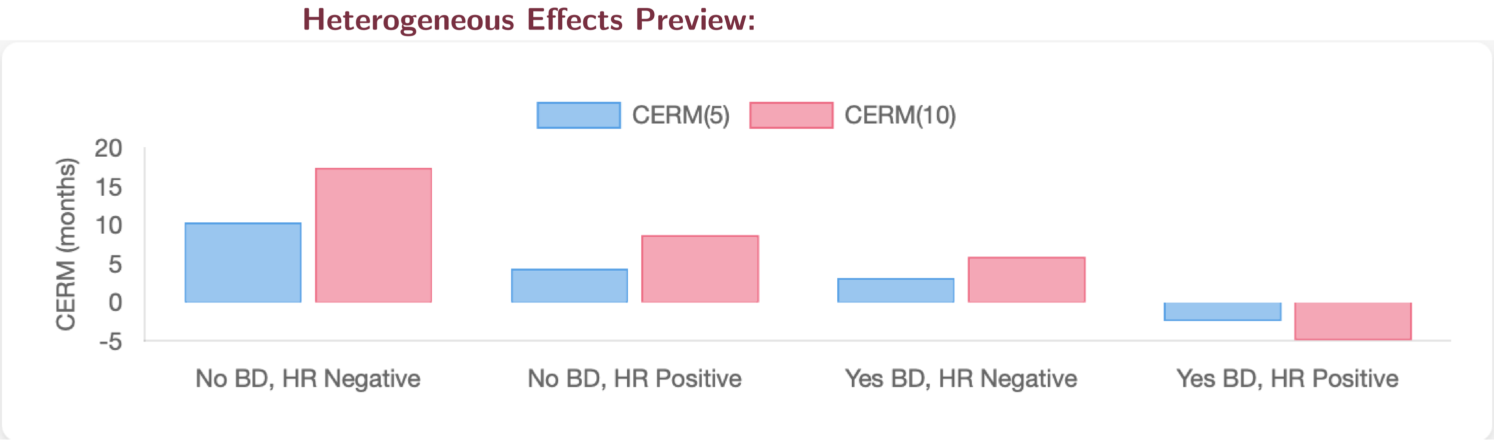
Key Estimand — CERM:

$$\boxed{\mathrm{CERM}_{ij}(t^*) = \mathbb{E}\left[\{T_{ij}(1) \wedge t^*\} - \{T_{ij}(0) \wedge t^*\} \,\middle|\, \mathbf{x}_{ij}, \text{County} = i\right]}$$

Expected gain in restricted mean survival: short vs. long TD

Assumptions: SUTVA, Consistency, Unconfoundedness, Positivity, Non-informative censoring

Heterogeneous Effects Preview:



Average CERM (months) by Biopsy Delay and HR status at 5 & 10 years — effects vary substantially across patient subgroups

Proposed Method: PS-LND Framework

Data $(y, \delta, z, \mathbf{x})$ $\longrightarrow$ **Stage 1:** PS $\hat{e}_{ij}$ $\longrightarrow$ **Stage 2:** Outcome $\longrightarrow$ **Causal Effects**

Stage 1: Propensity Score (Spatial Probit-SBART)

$$P(Z_{ij} = 1 \mid \mathbf{x}_{ij}, V_i) = \Phi\Big(\underbrace{b(\mathbf{x}_{ij})}_{\text{SBART}} + \underbrace{V_i}_{\text{DAGAR}}\Big)$$

$b(\cdot)$: unknown function — **Soft BART prior** (smooth sigmoid splits)
$\mathbf{V} \sim N(\mathbf{0}, \tau_v^{-1}\mathbf{Q}(\rho_v)^{-1})$: **DAGAR prior** for spatial confounding
Output: $\hat{e}_{ij} = E[\Phi\{b(\mathbf{x}_{ij}) + V_i\} \mid \mathcal{D}_1]$

Stage 2: Outcome Model (Log-Normal AFT + BCF)

$$\log T_{ij} \sim N\Big(\underbrace{\mu(\mathbf{x}_{ij}, \hat{e}_{ij})}_{\text{prognostic SBART}} + \underbrace{\tau(\mathbf{x}_{ij})}_{\text{treatment SBART}} \cdot Z_{ij} + \underbrace{W_i}_{\text{DAGAR}},\ \sigma^2\Big)$$

$\mu(\cdot)$: baseline survival (includes $\hat{e}_{ij}$ for targeted selection)
$\tau(\cdot)$: heterogeneous treatment effect
$\mathbf{W} \sim N(\mathbf{0}, \tau_w^{-1}\mathbf{Q}(\rho_w)^{-1})$: county-level spatial effects
Censored y_{ij}: data augmentation $\to$ sample latent $\tilde{y}_{ij}$

Why SBART?

Smooth sigmoid splits
No pre-specified interactions
Proven for causal inference

Why DAGAR over CAR/ICAR?

Proper joint distribution
Positive-definite precision
Interpretable $\rho_w \in (0, 1)$
Scalable to many clusters

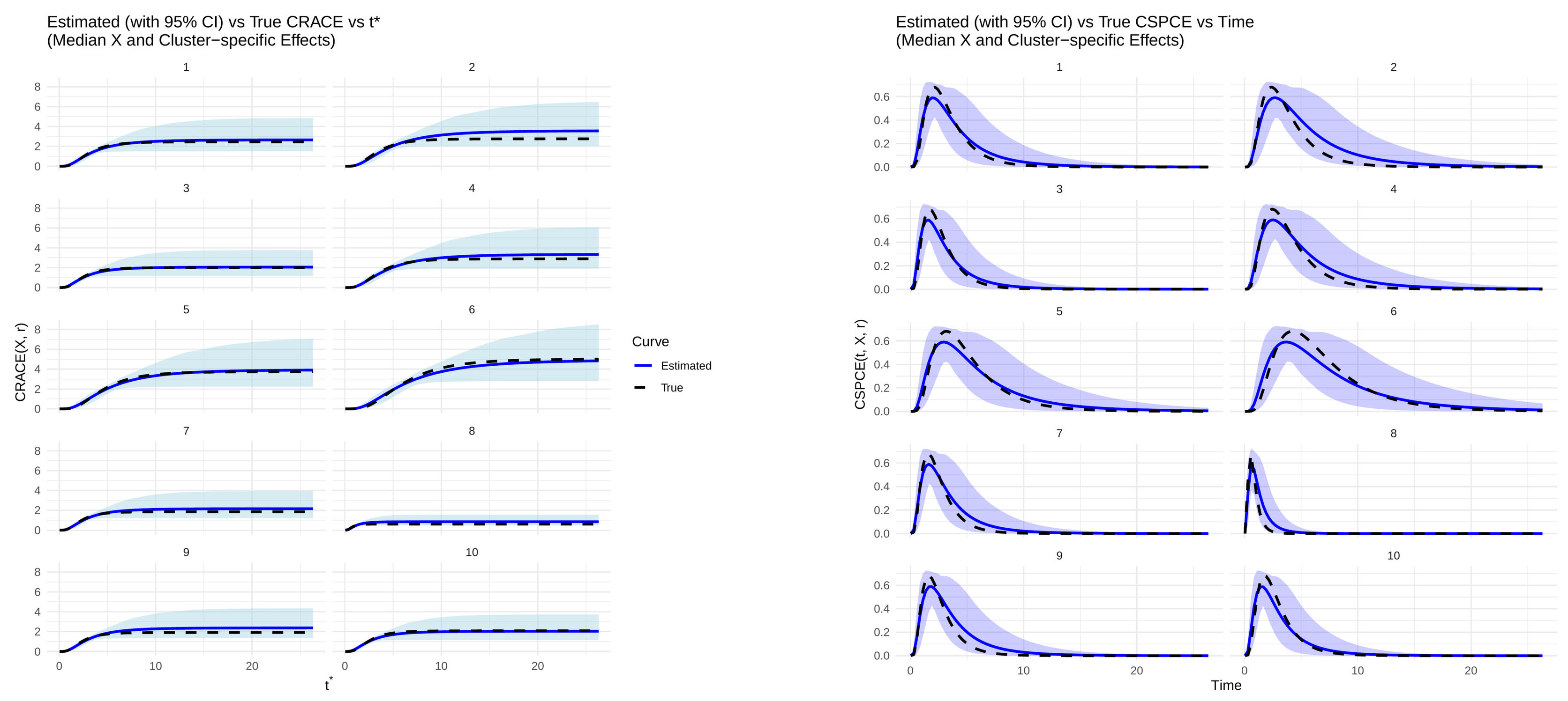
Aggregation: $\widehat{\mathrm{ACERM}}_i = \frac{1}{n_i}\sum_j \widehat{\mathrm{CERM}}_{ij}$; $\widehat{\mathrm{RATE}} = \frac{1}{N}\sum_{i,j} \widehat{\mathrm{CERM}}_{ij}$

Simulation Results

$K = 35$ clusters on 5×7 grid, $N = 1750$, 5 confounders, 150 replications

Scenario	Metric	PS-LND	riAFTBART	PSW-Fr.	IPW
1: Correct	Cov / MAE	**0.93–0.97 / 0.01**	0.46–0.95 / 0.08–0.58	0.54–0.97 / 0.08–0.50	0.38–0.52 / 0.06
2: Misspec.	Cov / MAE	**0.92–0.95 / 0.01**	0.77–0.89 / 0.20–0.79	0.64–0.89 / 0.21–0.63	0.42–0.53 / 0.06
3: Weibull	Cov / MAE	**0.87–0.90 / 0.03**	0.55–0.68 / 0.24–0.66	0.76–0.82 / 0.08–0.21	0.42–0.47 / 0.53

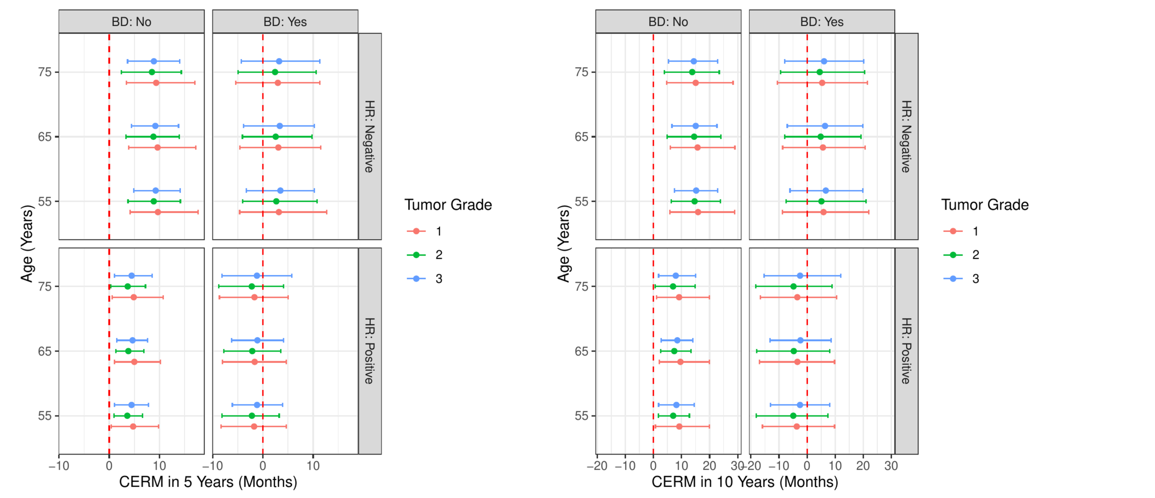
PS-LND: best coverage & lowest MAE — robust under misspecification & distributional violations



Est. (blue, 95% CI) vs. true (dashed) causal effects across 10 clusters (Scenario 1)

Application: Florida Cancer Registry — WA-Distant Stratum

Patient-Level Effects (Broward County)



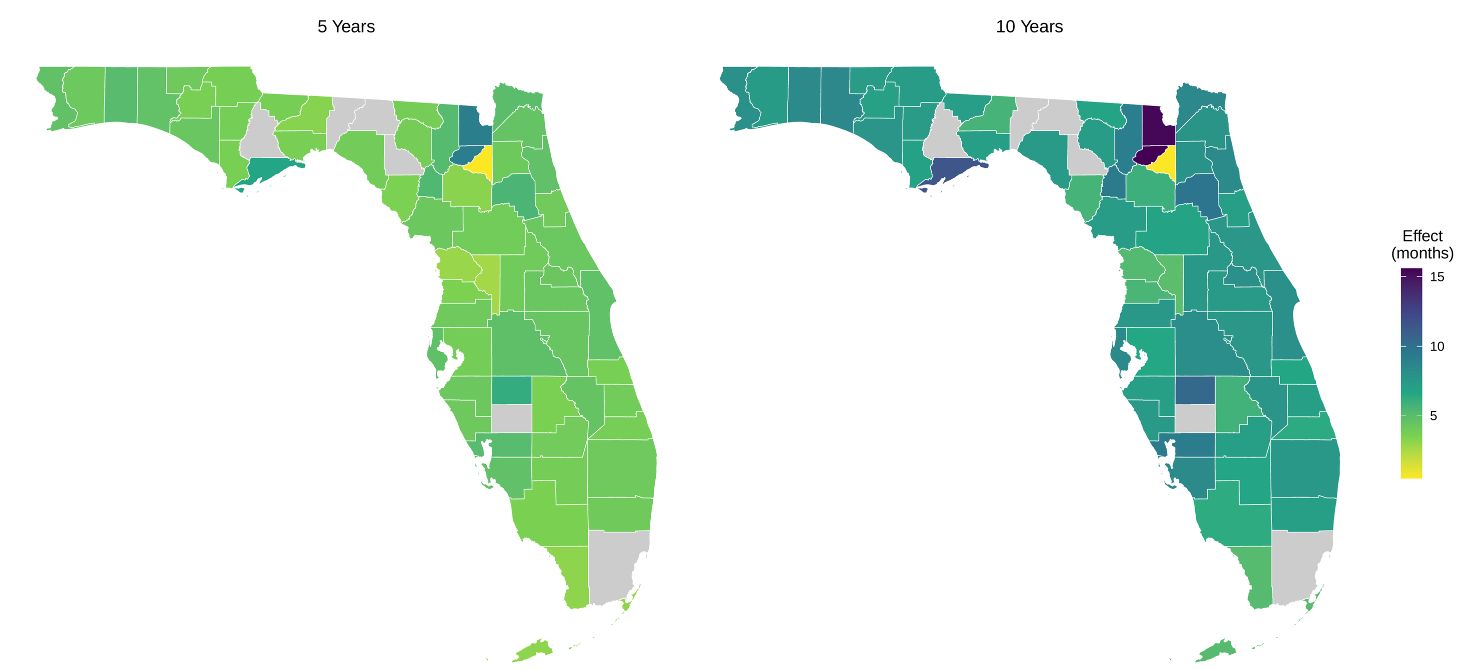
(a) 5-year CERM estimates. (b) 10-year CERM estimates.

Fig 1: Point and 95% interval estimates of CERM (in months) restricted within intervals of (a) 5 and (b) 10 years for WA patients with Distant-stage cancer in Broward County

$\widehat{\mathrm{CERM}}$ (months) at 5 & 10 yr by Age, BD, HR, TG ($n = 183$)

HR−, no BD: **13.8–15.9 mo.** gain (10 yr)
HR+, no BD: **6.9–9.6 mo.** gain
With BD: no significant effect (CI $\ni$ 0)

County-Level Spatial Patterns



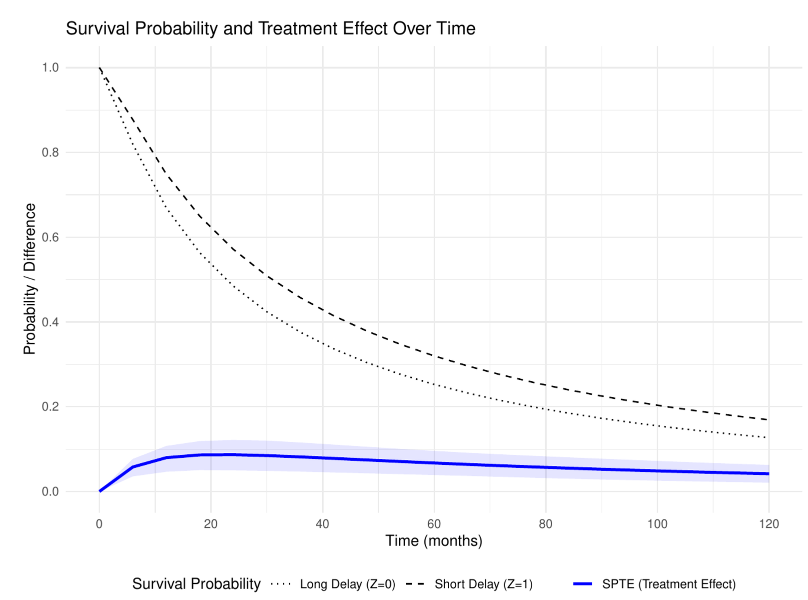
$\widehat{\mathrm{ACERM}}_i$ at 5 (L) & 10 (R) yr. Yellow = 0 mo → dark = 15+ mo; gray = no data

Highest: Union (>15 mo.), Baker (>13 mo.)
Near-zero: Bradford (CI $\ni$ 0)
Clear **spatial clustering** of effects

State-Wide Treatment Effects

Estimand	N	Est. (mo.)	95% CI
RATE (10 yr)	1994	**7.62**	(4.42, 10.86)
RATT (10 yr)	687	6.74	(3.09, 10.08)
RATU (10 yr)	1307	**8.09**	(4.60, 11.42)
RATE (5 yr)	1994	4.43	(2.56, 6.23)

No CI contains zero ⇒ strong evidence
RATU > **RATT**: long-TD patients gain the most
Benefits persist beyond 5 years



$\widehat{\mathrm{SPTE}}(t)$: peak **11%** survival gain at $t = 2.2$ yr; CI > 0 throughout

Policy Takeaway

Short TD ⇒ **7.6 mo.** average survival gain
Priority: **HR− patients without biopsy delay**
Target **rural northern counties** for navigation programs

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